60-DAY DATA ENGINEER ROADMAP

Detailed Weekly Learning Guide with Sources

Master SQL, Data Modeling, Databricks & System Design

Target Role	Duration	Daily Commitment	Total Hours	GitHub Repos
Senior Data Engineer (UAE)	60 Days (8 Weeks)	4 Hours/Day	240 Hours	7+ Repos

■ Primary Learning Resources (Your Subscriptions)

Databricks Academy: https://academy.databricks.com/

LinkedIn Learning: https://linkedin.com/learning

YouTube: Various data engineering channels

■ Supplementary Free Resources

LeetCode Database: https://leetcode.com/problemset/database/ (75 problems)

SQLZoo: https://sqlzoo.net/ (Interactive SQL)

Mode Analytics: https://mode.com/sql-tutorial/ (SQL tutorials)

Kimball Group: https://www.kimballgroup.com/ (Data warehouse design)

System Design Primer: https://github.com/donnemartin/system-design-primer

WEEK 1: SQL Fundamentals & Basics (Days 1-7)

■ Week 1 Goals

✓ Master basic SQL: Joins, Aggregations, Group By, Having
✓ Solve 25 SQL problems
✓ Build strong foundation for interviews
✓ Start GitHub portfolio (sql-mastery repo)

DAY 1: Joins & Multiple Tables

Topics:

INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, Cross joins
Join conditions and filtering, Multiple table joins (3+ tables)

Learning Sources (Total: 90 mins):

1. YouTube: "SQL Joins Explained" (15 mins)

Watch: KhanAcademy or Alex The Analyst version

2. LinkedIn Learning: SQL Essential Training (45 mins)

Navigate to: Course → SQL Training → Joins Section

3. SQLZoo Interactive Tutorial (30 mins)

Complete: "SELECT from multiple tables" section

Hands-on Practice (60 mins):

1. Find all customers and their orders (LEFT JOIN)
2. Find customers who have NOT made any order
3. Join 3+ tables together

■ Deliverables:

[ ] 5 JOIN queries written and tested
[ ] GitHub Commit: 'feat: SQL joins - 5 practice queries'

DAY 2: Aggregation Functions & GROUP BY

Topics:

COUNT, SUM, AVG, MIN, MAX | GROUP BY clause | HAVING clause vs WHERE
COUNT(DISTINCT) | Aggregate without GROUP BY

Learning Sources (90 mins):

YouTube: 'SQL Aggregation Functions' (20 mins)

LinkedIn Learning: Aggregation module (40 mins)

LeetCode SQL: 5 EASY GROUP BY problems (30 mins)

Hands-on Practice (60 mins):

1. Count total orders per customer
2. Sum revenue by product category
3. Find average order value per month
4. Find categories with >5 products (HAVING)

■ Deliverables:

[ ] 6 GROUP BY/HAVING queries

[ ] 5 LeetCode problems solved

DAY 3: Window Functions - ROW_NUMBER, RANK (60 mins learning + 60 mins practice)

YouTube: 'Window Functions Explained' | LinkedIn: Advanced SQL module | Mode: Window Functions

Deliverable: 5 window function queries | GitHub Commit

DAY 4: LAG, LEAD, Running Totals (60 mins learning + 60 mins practice)

Focus: Month-over-month growth, consecutive sequences, running averages

Deliverable: 5 advanced window queries | 5 LeetCode MEDIUM problems

DAY 5: CTEs & Subqueries (60 mins learning + 60 mins practice)

Topics: WITH clauses, multiple CTEs, recursive CTEs, query optimization

Deliverable: 5 CTE queries | Subquery vs CTE comparison

DAY 6: Cohort & Funnel Analysis (60 mins learning + 60 mins practice)

Real-world: Retention heatmap, funnel analysis, cohort retention

Deliverable: Complete cohort analysis | Funnel metrics | GitHub Commit

DAY 7: Week 1 Review & Interview Prep

[ ] Solve 10 LeetCode problems (total: 25+)

[ ] Complete 3 mock interviews

[ ] Create sql-mastery GitHub repository

[ ] Commit: 'docs: SQL mastery week 1 complete'

■ Week 1 Summary:

■ Joins, Aggregations, Window Functions, CTEs mastered

■ 25+ SQL problems solved

■ sql-mastery repository created

WEEK 2: SQL Advanced & DBMS Fundamentals (Days 8-14)

■ Week 2 Goals

✓ Complete 50 total SQL problems (25 → 50)
✓ Master DBMS: ACID, Indexing, Query Plans, Partitioning
✓ Create 10-page DBMS notes document
✓ Continue interview practice

Day	Topic	Duration	Key Resource	Deliverable
8	Recursive CTEs	2 hours	YouTube + LinkedIn	5 recursive queries
9	String/Date Functions	2 hours	YouTube + Mode	6 function queries
10	ACID Properties	2 hours	YouTube + PostgreSQL Docs	ACID document
11	Isolation Levels	2 hours	YouTube + PostgreSQL Docs	Isolation comparison
12	Indexing Strategies	2 hours	YouTube + LinkedIn	Index analysis + guide
13	Partitioning	2 hours	YouTube + PostgreSQL Docs	Partitioning strategy
14	Week 2 Review	4 hours	All resources	Complete DBMS guide

Key DBMS Concepts to Master:

ACID Properties: Atomicity, Consistency, Isolation, Durability with examples
Isolation Levels: READ UNCOMMITTED, COMMITTED, REPEATABLE READ, SERIALIZABLE
Query Execution: EXPLAIN ANALYZE, cost estimation, optimization
Indexing: B-tree, Hash, Composite indexes, when to use each
Partitioning: Range, list, hash partitioning for performance

■ Week 2 Summary:

■ SQL: 25 → 50 LeetCode problems (halfway!)
■ DBMS: All concepts mastered with practical examples
■ Created: 10-page DBMS fundamentals guide
■ GitHub: dbms-fundamentals repo created

WEEK 3: Data Modeling & Warehousing (Days 15-21)

■ Week 3 Goals

✓ Master star schema design
✓ Learn SCD Type 1, 2, 3
✓ Design 3 warehouses: Amazon, Netflix, Uber
✓ Create professional architecture diagrams

Learning Path (3 stages):
Stage 1 (Days 15-17): Concepts
→ YouTube: Star Schema Explained (25 mins)
→ LinkedIn Learning: Dimensional Modeling (50 mins)
→ Kimball Group Resources (20 mins)

Stage 2 (Days 18-20): 3 Real-World Case Studies
→ Amazon Analytics Warehouse
→ Netflix Analytics Warehouse
→ Uber Analytics Warehouse

Stage 3 (Day 21): Intro to Data Vault
→ Hubs, Links, Satellites
→ Data Vault vs Star Schema comparison

Case Study Structure (Days 18-20):

For each company: 1. Design ERD **|** 2. Write DDL **|** 3. Create sample data **|** 4. Document design decisions

■ Week 3 Summary:

■ 3 professional ERD diagrams created
■ 3 complete SQL DDL scripts
■ data-warehouse-designs repo with all designs

WEEKS 5-6: Databricks & PySpark (Days 29-42)

Primary Resource: Databricks Academy

Link: https://academy.databricks.com/

Duration: 20 hours across both weeks

WEEK 5 (Days 29-35): Spark Fundamentals & Delta Lake
• Days 29-30: RDDs, lazy evaluation, transformations vs actions
• Days 31-32: DataFrames, Spark SQL, optimization
• Days 33-34: Delta Lake (ACID, time travel, merge)
• Day 35: Retail Analytics Pipeline capstone (Bronze → Silver → Gold)

WEEK 6 (Days 36-42): PySpark Optimization
• Days 36-37: Partitioning, shuffle, broadcast joins
• Days 38-39: Performance tuning, execution plans, memory management
• Days 40-41: Advanced features (window functions, UDFs)
• Day 42: Optimize slow job (30 min → 5 min target)

■ Medallion Architecture Implementation:

Bronze Layer: Raw data ingestion as-is

Silver Layer: Cleaned, deduplicated, validated data

Gold Layer: Aggregations and business-ready tables

WEEK 7: Kafka & Event Streaming (Days 43-49)

■ Week 7 Goals

✓ Master Kafka architecture and patterns
✓ Design 3 system architectures
✓ Create professional architecture diagrams

Days 43-44: Kafka Fundamentals
• Topics, partitions, consumer groups, offset management
• Producer/consumer patterns
• Kafka Connect basics
Resources: YouTube + Kafka Official Docs (8 hours)

Day 45-46: Kafka Architecture
• Event streaming patterns
• Exactly-once semantics
• Streaming joins and enrichment

Days 47-49: System Design Projects
1. Clickstream Analytics Platform
2. Customer 360 Platform
3. Fraud Detection Pipeline

WEEK 8: Airflow, GenAI & Final (Days 50-60)

Days 50-51: Apache Airflow
• DAG concepts and task dependencies
• Scheduling, monitoring, error handling
• Resource: YouTube + Airflow Docs (6 hours)

Day 52: System Design Review
• Refine all 3 architectures
• Complete data flow diagrams
• Technical specifications

Days 53-55: GenAI & RAG Intro
• LLM basics and RAG concepts
• Build Data Engineering Copilot (MVP)
• Resource: LangChain Docs + YouTube (14 hours)

Days 56-60: Portfolio Finalization
• Polish all 7+ GitHub repositories
• Write comprehensive READMEs
• Complete documentation
• Final mock interviews

■ Complete Resource Summary Table

Topic	Primary Resource	Duration	Link/Notes
SQL	LeetCode Database	30 hrs	https://leetcode.com/problemset/database/
	SQLZoo	5 hrs	https://sqlzoo.net/ - Interactive
	Mode Analytics	5 hrs	https://mode.com/sql-tutorial/ - Tutorials
DBMS	LinkedIn Learning	8 hrs	Course modules on ACID, Indexing, Partitioning
	YouTube Channels	5 hrs	Search: DBMS fundamentals, query optimization
Data Modeling	Kimball Group	6 hrs	https://www.kimballgroup.com/ - Articles
	YouTube	4 hrs	Star schema, data vault tutorials
Data Vault	LinkedIn Learning	6 hrs	Data Vault 2.0 module
Databricks/Spark	Databricks Academy	20 hrs	https://academy.databricks.com/ - FREE
Kafka	YouTube	8 hrs	Search: Apache Kafka tutorial series
	Official Docs	4 hrs	https://kafka.apache.org/documentation/
Airflow	YouTube	6 hrs	Search: Apache Airflow DAG tutorial
System Design	System Design Primer	10 hrs	https://github.com/donnemartin/system-design-primer
GenAI/RAG	LangChain Docs	8 hrs	https://python.langchain.com/ - Free docs
	YouTube (ArjanCodes)	6 hrs	https://youtube.com/@ArjanCodes - AI Architecture

■ Daily Schedule Template

Hour 1 (60 mins) - LEARNING
• Watch video tutorial or read documentation
• Take notes on key concepts
• Example: "SQL Joins" from YouTube + LinkedIn Learning

Hour 2 (60 mins) - HANDS-ON PRACTICE
• Write code/SQL queries based on what you learned
• Complete interactive exercises (SQLZoo, LeetCode)
• Debug and test your code

Hour 3 (60 mins) - ARCHITECTURE/DESIGN
• Draw diagrams (ERD, system design, data flow)
• Design solutions to case studies
• Plan query optimization or architecture

Hour 4 (60 mins) - PROJECT WORK
• Push code to GitHub
• Update documentation
• Commit with meaningful message
• Track progress and reflect

■ Success Metrics: By Day 60

SQL & DBMS

- ■ Write complex queries with 5+ JOINs
- ■ Explain ACID properties with examples
- ■ Design indexing strategy for 1B row table
- ■ Solve 75 LeetCode database problems

Data Warehousing

- ■ Design warehouse for any company
- ■ Choose between Star, Snowflake, Data Vault, Medallion
- ■ Explain SCD types with implementations
- ■ Create professional architecture diagrams

Spark & Delta

- ■ Optimize Spark job (30 mins → 5 mins)
- ■ Explain Delta Lake ACID guarantees
- ■ Design Medallion pipeline (Bronze → Silver → Gold)
- ■ Use Spark SQL and PySpark APIs fluently

Kafka & System Design

■ Design real-time streaming architecture
■ Build Airflow DAG with proper error handling
■ Draw complete system architecture
■ Explain exactly-once processing semantics

GenAI & RAG

■ Explain RAG vs fine-tuning vs in-context learning
■ Build basic RAG system with embeddings
■ Create Data Engineering Copilot demo
■ Discuss LLM integration in data pipelines

Portfolio

■ 7+ GitHub repositories with 100+ commits
■ Comprehensive READMEs with examples
■ Professional documentation
■ Clean, production-ready code

This is an ambitious but achievable 60-day plan.

Success depends on:
1. Consistent daily effort (4 hours every day)
2. Building projects (not just watching videos)
3. Pushing code to GitHub daily (visible proof)
4. Staying focused on high-ROI topics
5. Regular weekly review and progress tracking

Good luck! You got this! ■